

Solutions second chapter the Standard model and Beyond

Abstract

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Useful formulas

A list of formulas used throughout the exercises.

$$s = E_{\text{CM}}^2 = 4(p^2 + m^2)$$

$$t = -2p^2(1 - \cos(\theta))$$

$$u = -2p^2(1 + \cos(\theta))$$

$$\phi_0 = \int [a(\vec{p})e^{-p \cdot x} + a^\dagger(\vec{p})e^{+ip \cdot x}] \frac{d^3\vec{p}}{(2\pi)^3 2E_p}$$

1 Exercise 1

The hamiltonian for a free complex scalar field is

$$H = \int \left[\left(\frac{\partial \phi^\dagger}{\partial t} \right) \left(\frac{\partial \phi}{\partial t} \right) + (\vec{\nabla} \phi^\dagger) \cdot (\vec{\nabla} \phi) + m^2 \phi^\dagger \phi \right] d^3 \vec{x}.$$

Show that

$$H = \int E_p \left[a^\dagger(\vec{p}) a(\vec{p}) + b^\dagger(\vec{p}) b(\vec{p}) \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p}$$

Assume that H is normal ordered.

To solve this exercise, we just need to compute everything using the field expression in function of the creators and annihilations operators. An example of such field is the ϕ_{π^+} which we take to annihilate a π^+ particle or, equivalently, to create a π^- . This means that $\phi_{\pi^+} = \phi_{\pi^-}^\dagger$.¹ With all this, the explicit definition of the fields are

$$\begin{aligned} \phi &= \int \left[a(\vec{p}) e^{-ip \cdot x} + b^\dagger(\vec{p}) e^{ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \\ \phi^\dagger &= \int \left[b(\vec{p}) e^{-ip \cdot x} + a^\dagger(\vec{p}) e^{ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \end{aligned}$$

Since we can go from ϕ to $\phi^\dagger$ by the substitution $a \mapsto b$, we compute the relevant quantities only for ϕ . Just so that we may speed up our calculations,

$$\begin{aligned} \frac{d}{dt} e^{-ip \cdot x} &= -iE_p e^{-ip \cdot x} \\ \frac{d}{dt} e^{ip \cdot x} &= iE_p e^{ip \cdot x} \\ \vec{\nabla} e^{-ip \cdot x} &= i\vec{p} e^{-ip \cdot x} \\ \vec{\nabla} e^{ip \cdot x} &= -i\vec{p} e^{ip \cdot x} \end{aligned}$$

Onwards!

¹It is in this sense that we might say that creating a particle and destroying an antiparticle is the same operation, although the two have opposite 4-momenta.

$$\begin{aligned}
\frac{d\phi}{dt} &= \int \left[a \frac{d}{dt} e^{-ip \cdot x} + b^\dagger \frac{d}{dt} e^{+ip \cdot x} \right] \frac{d^3 \vec{p}}{(2\pi)^3 2E} = \int [-iE a e^{-ip \cdot x} + iE b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{d(2\pi)^3 2E_p} \\
&= \int iE [-a e^{-ip \cdot x} + b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{d(2\pi)^3 2E} \\
\vec{\nabla} \phi &= \int [i p a e^{-ip \cdot x} - i \vec{p} b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} = \int i \vec{p} [a e^{-ip \cdot x} - b^\dagger e^{+ip \cdot x}] \frac{d^3 \vec{p}}{(2\pi)^3 2E}
\end{aligned}$$

Therefore, by multiplying the correspond quantites of $\frac{d\phi^\dagger}{dt}$ and $\nabla \phi^\dagger$ we obtain

$$\begin{aligned}
H &= \int \left[\left(\frac{\partial \phi^\dagger}{\partial t} \right) \left(\frac{\partial \phi}{\partial t} \right) + (\vec{\nabla} \phi^\dagger) \cdot (\vec{\nabla} \phi) + m^2 \phi^\dagger \phi \right] d^3 \vec{x} \\
&= \int - (EE' + |\vec{p}| |\vec{p}'|) [a b e^{-2(\dots)} - a^\dagger a - b^\dagger b + a^\dagger b^\dagger e^{+2(\dots)}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} d^3 \vec{x} \\
&\quad + \int m^2 [a b e^{-2(\dots)} + a^\dagger a + b^\dagger b + b^\dagger a^\dagger e^{+2(\dots)}] \frac{d^3 \vec{p}}{(2\pi)^3 2E} \frac{d^3 \vec{p}'}{(2\pi)^3 2E} d^3 \vec{x}
\end{aligned}$$

where $(\dots) = i(p + p') \cdot x$. These are a lot of terms, so something ought to cancel. Remember that we still have an integration to carry out, and this is explicitly

$$\int e^{\pm i(p+p') \cdot x} d^3 \vec{x} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}')$$

This means that we need to put $\vec{p} = \vec{p}'$. The last ingredient is using the on-mass shell prescription, which tells that we need to consider only the 4-momenta with $p^2 = m^2$.

The only terms which survive are those without an $e^{(\dots)}$, that is

$$H = \int 2E^2 [a^\dagger a + b^\dagger b] \frac{d^3 \vec{p}}{(2\pi)^3 2E^2} = \int E [a^\dagger a + b^\dagger b] \frac{d^3 \vec{p}}{(2\pi)^3 2E}$$

Now, let's see if this result make physically sense. First of all, we can count the mass-powers in the result and see that the whole has $[H]_m = 1$, which is good since it has to be an energy. Tracing back the definitions, remebering that $[\phi]_m = 1$, we have that the operator a, b have mass dimension of -1 , thereby showing that $[H]_m = 1$.

2 Exercise 2

The CM differential and total cross sections for the elastic scattering of a Hermitian scalar of mass m with potential $V_I(\phi) = k \frac{\phi^3}{3!} + \lambda \frac{\phi^4}{4!}$ are given by the formula

$$\frac{d\sigma}{d\cos(\theta)} = \frac{1}{32\pi s} \left| \lambda + k^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] \right|^2$$
$$\sigma = \frac{1}{2} \int_{-1}^1 \frac{d\sigma}{d\cos(\theta)} d\cos(\theta)$$

Choose $m = 0.5\text{GeV}$, $\lambda = 0.3$ and $k = 1.2\text{GeV}$. Plot the CM differential cross section $\frac{d\sigma}{d\cos(\theta)}$ in units of $\text{fm}^2 = 10^{-26}\text{cm}^2$ as a function of $\cos(\theta)$ for $s = 5m^2, 6m^2$, and $7m^2$, and plot the total cross section σ as a function of $\sqrt{s}$ for $2m < \sqrt{s} < 3m$.

First of all, we choose s and $\cos(\theta)$ as independent variables, so that

$$s = 4(p^2 + m^2)$$
$$t = -2 \left(\frac{2}{4} - m^2 \right) (1 - \cos(\theta))$$
$$u = -2 \left(\frac{2}{4} - m^2 \right) (1 + \cos(\theta))$$

3 Exercise 3

Derive the equation

$$\frac{d\sigma}{d\cos(\theta_3)} \stackrel{!}{=} \frac{|M_{fi}|^2}{32\pi p_1 m_2} \frac{p_3}{E_1 + m_2 - \frac{p_1 E_3}{p_3} \cos(\theta_3)}$$

, where the θ_3 refers to the laboratory scattering angle of particle 3, for the special case $m_1 = m_3 = 0$ and $m_2 = m_4$ by Lorentz transforming the equation

$$\frac{d\sigma}{d\cos(\theta)} = \frac{1}{32\pi s} \frac{p_f}{p_i} |M_{fi}|^2$$

We may use the fact that σ, s and t are Lorentz's invariants.

The exercise asks to go from the center of momentum frame (or center of mass) straight into the laboratory frame. Therefore, we need to check how $\cos(\theta), p_i$ and p_f change as we go from one coordinate system to another.

So, let's write what we know about the CM frame. Knowing that in this frame the total three momentum cancels (hence the name), we have

$$\begin{aligned} p_1 &= (E_1 \quad \vec{p}_i) \\ p_2 &= (E_2 \quad -\vec{p}_i) \end{aligned}$$

Now, to go from this frame to the lab one, that is to

$$\begin{aligned} p_1 &= (E_1^{\text{lab}} \quad \vec{p}_1) \\ p_2 &= (E_2^{\text{Lab}} \quad \vec{p}_2) \end{aligned}$$

we use the relativistic Lorentz addition imposing $\vec{\beta} = \frac{\vec{p}_i}{E_2}$. With this requirement, we get that $\vec{p}_2 = \vec{0}$, since²

$$\begin{aligned} \vec{p}_2 &= \gamma_L(-\vec{p}_i + \vec{\beta}_L E_2) \\ &= \gamma_L(-\vec{p}_i + \vec{p}_i) = 0 \end{aligned}$$

Without further ado, let's compute what are the 4-momentas p_1 and p_3 in the new frame.

²In this example, I'm tacitly assuming that all the computations are in collinear frames, that is $\vec{p}_i = (|\vec{p}_i| \quad 0)$

$$\begin{aligned}
E_1^{\text{Lab}} &= \frac{E_2}{m_2} \left(E_1 + \frac{1}{E_2} |\vec{p}_i|^2 \right) \\
&= \frac{E_2 E_1}{m_2} + \frac{1}{m_2} (E_2^2 - m_2^2) = \frac{E_2 \sqrt{s}}{m_2} - m_2 \\
p_1^{\vec{\text{Lab}}} &= \frac{E_2}{m_2} \left(1 + \frac{E_1}{E_2} \right) \vec{p}_i = \frac{E_1 + E_2}{m_2} \vec{p}_i = \frac{\sqrt{s}}{m_2} \vec{p}_i \\
E_3^{\text{Lab}} &= \frac{E_2 E_3}{m_2} + \frac{1}{m_2} |\vec{p}_i| |\vec{p}_f| \cos(\theta) \\
\vec{p}_3^{\text{Lab}} &= \left(\frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i| \right) \vec{p}_i + |\vec{p}_f| \sin(\theta) \vec{p}_\perp
\end{aligned}$$

Since we have computed all the quantities in the new frame, I will call them $E_1, E_3, \vec{p}_1$ and $\vec{p}_3$ without the superscript "Lab". The last expression allows us to get how the new angle of scattering θ_3 is related to the old scattering angle θ . To see this, we name the components of $\vec{p}_3$ as $\vec{p}_3 = (|\vec{p}_3| \cos(\theta_3) \quad |\vec{p}_3| \sin(\theta_3))$ so that

$$\begin{cases} |\vec{p}_3| \cos(\theta_3) = \frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i| \\ |\vec{p}_3| \sin(\theta_3) = |\vec{p}_f| \sin(\theta) \end{cases}$$

Since the two expressions above are simultaneously valid, we can take the first equation and divide it by the second, getting

$$\tan(\theta_3(\theta)) = \frac{|\vec{p}_f| \sin(\theta)}{\frac{E_2}{m_2} |\vec{p}_f| \cos(\theta) + \frac{E_3}{m_2} |\vec{p}_i|}$$

Our objective is to compute the differential cross section in the new frame. Therefore, we need to change variables from $\theta \mapsto \theta_3$, that is

$$\frac{d\sigma}{d\cos(\theta_3)} \frac{d\cos(\theta_3)}{d\cos(\theta)}$$

Luckily, our work from before may be used precisely for this. First of all, we have $\tan(\theta_3)$ and not $\cos(\theta_3)$ as a function of the old variables. But, we can employ a trigonometric trick to rewrite³

$$\cos(\theta_3) = \frac{\cos(\theta_3)}{\sqrt{\cos^2(\theta_3) + \sin^2(\theta_3)}} = \frac{1}{\sqrt{1 + \tan^2(\theta_3)}}.$$

³Here we are interested only in the magnitude of the derivative, so we may as well suppose that $\cos(\theta_3)$ is positive.

Furthermore,

$$\begin{aligned}
\frac{d\cos(\theta_3)}{d\cos(\theta)} &= \frac{d\theta}{d\cos(\theta)} \frac{d\cos(\theta_3)}{d\theta} = -\frac{1}{\sin(\theta)} \frac{d\cos(\theta_3)}{d\theta} \\
&= -\frac{1}{\sin(\theta)} \frac{d}{d\theta} \left(\frac{1}{\sqrt{1 + \tan^2(\theta_3(\theta))}} \right) \\
&= \frac{1}{\sin(\theta)} \frac{\tan(\theta_3)}{\sqrt{1 + \tan^2(\theta_3)}} \\
&= \frac{\sin(\theta_3)}{\sin(\theta)} = \frac{|\vec{p}_f|}{|\vec{p}_3|}.
\end{aligned}$$

With all this, we just turn to the differential cross section⁴,

$$\frac{d\sigma}{d\cos(\theta_3)} = \frac{d\sigma}{d\cos(\theta)} \frac{d\cos(\theta)}{d\cos(\theta_3)} = \frac{1}{32\pi |\vec{p}_1| m_2} \frac{|M_{fi}|^2}{\sqrt{s}}$$

Now, we just need to specialize to the case prescribed, that is where $m_1 = 0 = m_3$ and $m_2 = m_4$. From our general formula, we substitute $E_3 = |\vec{p}_3|$, $E_1 = |\vec{p}_1|$. Furthermore, we know that in this special case

$$\frac{p_1}{p_3} = \frac{1}{1 + \frac{p_1}{m_2}(1 - \cos(\theta_3))}$$

$$\begin{aligned}
\frac{d\sigma}{d\cos(\theta_3)} &\stackrel{\text{Special case}}{=} \frac{|M_{fi}|^2}{32\pi p_1 m_2} \frac{p_3}{|\vec{p}_1| + m_2 - |\vec{p}_1| \cos(\theta_3)} \\
&= \frac{|M_{fi}|^2}{32\pi |\vec{p}_1| m_2} \frac{p_3}{p_1 m_2 \left(1 + \frac{|\vec{p}_1|}{m_2} \cos(\theta_3)\right)} \\
&= \frac{|M_{fi}|^2}{32\pi m_2^2} \left(\frac{|\vec{p}_3|}{|\vec{p}_1|} \right)^2
\end{aligned}$$

⁴I currently lack the proof that $\sqrt{s} = E_1 + m_2 - \frac{|\vec{p}_1| E_3}{|\vec{p}_3|} \cos(\theta_3)$.

4 Exercise 4

Consider the process $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$ with $p_1 \neq p_3$ and $p_2 \neq p_4$, in the theory of a complex scalar field with Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - V_I(\phi, \phi^\dagger) = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - \frac{\lambda}{4} (\phi^\dagger \phi)^2.$$

The tree-level amplitude M_{fi} is

$$(2\pi)^4 \delta^4(p_3 + p_4 - p_1 - p_2) M_{fi} = \int \langle p_3 p_4 | -iV_I(\phi_0(x), \phi_0^\dagger(x)) | p_1 p_2 \rangle d^4x$$

Calculate the amplitude using the free-field expression for ϕ_0 , and show that $M_{fi} = -i\lambda$

Before diving into this exercise, let's see what are the states we are working with. The process we describe is $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$. As we have done back in exercise 1, let's decide that ϕ_π destroys π^+ particles while $\phi^\dagger$ destroys them. According to this convention, we have

$$\begin{aligned} a^\dagger(\vec{p}) |0\rangle &= |\pi^+(\vec{p})\rangle \\ b^\dagger(\vec{p}) |0\rangle &= |\pi^-(\vec{p})\rangle \end{aligned}$$

Then, we see that

$$|\vec{p}_1 \vec{p}_2\rangle = |\pi^+(\vec{p}_1)\pi^-(\vec{p}_2)\rangle = a^\dagger(\vec{p}_1)b^\dagger(\vec{p}_2) |0\rangle$$

Now, let's consider the product of these 4 fields. When we evaluate them, we get a combination of $a, b, a^\dagger, b^\dagger$ which either creates or destroy particles or antiparticles. Within these 16 different terms, what should we keep? The $\pi^\pm$ is a scalar field, so the final state can have multiple bosons, but here we are interested in the scalar product with another 2-particle state, that is $\langle \pi^+(\vec{p}_3)\pi^-(\vec{p}_4) |$. Therefore, the combination of $a, b, a^\dagger, b^\dagger$ that we keep are only those which leave the final number of particles intact, that is 1 particle and 1 antiparticle. To compactify the notation, call

$$\begin{aligned} A_{pq}^{lm} &\equiv \frac{1}{(2\pi)^{3(l+m+p+q)}} \frac{1}{(2E_{p_i})^l (2E_{p_j})^m (2E_{p_k})^p (2E_{p_n})^q} \times \\ &\quad (a^\dagger(\vec{p}_a))^l e^{+i\left(\sum_{k=1}^l p_k\right) \cdot x} a^m(\vec{p}_b) e^{-i\left(\sum_{k=1}^m p_k\right) \cdot x} b^p(\vec{p}_c) e^{-i\left(\sum_{k=1}^p p_k\right) \cdot x} (b^\dagger(\vec{p}_d))^q e^{+i\left(\sum_{k=1}^q p_k\right) \cdot x} \\ &= \frac{1}{(2\pi)^{3(l+m+p+q)}} \frac{1}{(2E_{p_i})^l (2E_{p_j})^m (2E_{p_k})^p (2E_{p_n})^q} \times \\ &\quad (a^\dagger(\vec{p}_a))^l a^m(\vec{p}_b) b^p(\vec{p}_c) (b^\dagger(\vec{p}_d))^q e^{+i\left(\sum_{i,j,k,n=1}^{l,m,p,q} (\vec{p}_i + \vec{p}_n - \vec{p}_j - \vec{p}_k) \cdot x\right)} \end{aligned}$$

The only A's which survive are A_{11}^{11} , A_{00}^{22} or A_{22}^{00} . We have only 1 way to get A_{00}^{22} and A_{22}^{00} , whereas we have 4 ways to write down A_{11}^{11} .

5 Exercise 5

Consider the following Lagrangian density

$$\mathcal{L} = \sum_{i=1}^3 \frac{1}{2} [(\partial_\mu \phi_i)^2 - m_i^2 \phi_i^2] - k \phi_1 \phi_2 \phi_3$$

for three non-identical Hermitian fields. Calculate the lowest order differential cross section in the center of mass for $\phi_1(\vec{p}_1) \phi_2(\vec{p}_2) \rightarrow \phi_1(\vec{p}_3) \phi_2(\vec{p}_4)$ as a function of s and the CM scattering angle θ for the special case $m_1 = m_2 \neq m_3$.

6 Exercise 6

Suppose that the interaction potential for a complex scalar field were

$$V_I = \sigma_4(\phi^4 + \phi^{4,\dagger}),$$

(rather than $\lambda \frac{(\phi\phi^\dagger)^2}{4}$), which is not $U(1)$ invariant. Show that the charge is not conserved and calculate the lowest order amplitude for $\pi^+(\vec{p}_1)\pi^+(\vec{p}_2) \rightarrow \pi^-(\vec{p}_3)\pi^-(\vec{p}_4)$.

7 Exercise 7

Consider charged pion QED with a massive photon, i.e. add the term $\frac{1}{2}M_A^2 A_\mu A^\mu$ to the lagrangian density (insert qed lagrangian) (with $V_I = 0$). assume $M_A > 2m$.⁵

1. Calculate the decay rate for $\gamma \rightarrow \pi^+ \pi^-$ in the photon rest frame at tree level for an unpolarized massive photon.
 2. Calculate the π^+ angular distribution $\frac{d\Gamma}{d\cos(\theta)}$ for a polarized photon, where θ is the angle between the photon polarization direction in the rest frame and the π^+ direction.
 3. Show that one recovers the result in 1. when $\frac{d\Gamma}{d\cos(\theta)}$ is integrated over $\cos(\theta)$.
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⁵The photon mass term is not gauge invariant, but the model still makes sense at tree level

8 Exercise 8

Consider $\pi^+(\vec{p}_1)\pi^-(\vec{p}_2) \rightarrow \pi^+(\vec{p}_3)\pi^-(\vec{p}_4)$ in massive scalar electrodynamics, i.e. with

$$V_I = g\phi^\dagger,$$

where A , the analog of the electromagnetic field, is a Hermitian spin-0 field with mass $\mu \neq 0$. The analog of the charge is g , which now has dimension of mass.

1. Find expressions for the differential and total cross sections in the CM to lowest non-trivial order, in terms of s, m, μ, g and $\cos(\theta)$.
 2. Define the dimensionless variable $x = \frac{s}{m^2} \geq 4$, and specialize to the values $m = 1\text{GeV} = g$, $\mu = 0.5\text{GeV}$. Plot $\frac{d\sigma}{d\cos(\theta)}$ in units of $fm^2 = 10^{-26}\text{cm}^2$ for $x = 4, 4.2$ and 4.4 .
 3. Plot σ in units of fm^2 versus x for the same parameter values and the range $4 \leq x \leq 5$.
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9 Exercise 9

Prove directly from the defining relations that

$$\sigma^{\mu\nu}\gamma^5 = -\frac{i}{2}\epsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma}.$$

10 Exercise 10

Prove the Gordon decomposition formulas

$$\begin{aligned}2m(\bar{u}_2\gamma^\mu u_1) &= \bar{u}_2(p_1 + p_1)^\mu u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 - p_1)_\nu u_1 \\2m(\bar{u}_2\gamma^\mu\gamma^5 u_1) &= \bar{u}_2(p_2 - p_1)^\mu\gamma^5 u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 + p_1)_\nu\gamma^5 u_1 \\0 &= \bar{u}_2(p_2 - p_1)^\mu u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 + p_1)_\nu u_1 \\0 &= \bar{u}_2(p_1 + p_1)^\mu\gamma^5 u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_2 - p_1)_\nu\gamma^5 u_1\end{aligned}$$

where $u_{1,2}$ are two Dirac u spinors for a particle of mass m .
